

ELECTIVE-III

UME740 : Fracture Mechanics

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Course Objective: The aim of this course is to develop an understanding of the mechanics of fracture of engineering materials and structures under static and dynamic loading. This understanding is essential for the assessment of integrity and durability of structures and structural components in the presence of structural defects, so as to ensure reliability and safety.

Syllabus

Introduction: Introduction to the realm of fracture and back ground history of development of fracture mechanics; Discrepancy between theoretical and real strength of materials, conventional failure criteria based on stress concentration and characteristic brittle failures, Griffith's work.

Linear Elastic Fracture Mechanics (LEFM) Based Design Concepts: Crack deformation modes and basic concepts, crack tip stresses and deformation, stress intensity factor (SIF) and its criticality in different modes, superposition of SIFs, LEFM design concept applications; Concept of energy release rate, equivalence of energy release rate and SIF.

Fracture Toughness: Fracture toughness and its laboratory determination procedure, test specimen size requirement etc.; Effect of temperature and loading rate on fracture toughness; Fatigue and fatigue crack propagation laws, fatigue life calculations under constant and variable amplitude loading, mixed-mode fatigue crack propagation.

Strain Energy Density Failure Criterion: Introduction, volume strain energy density, basic hypothesis and application of energy density based failure criteria for two and three dimensional linear elastic crack problems.

Elastic Plastic Fracture Mechanics Based Design Criteria: Design criteria for non-brittle materials; plastic zone corrections, crack opening displacement (COD), J-contour integral and crack growth resistance (R-curve) concepts.

Research Assignment

Each team of 4-5 students will submit a case study on a fracture mechanics problem. Students are expected to enhance their knowledge, problem solving skills and understanding of the subject matter through completing the research assignment. The assignments are marked, with the mark contributing to the final grade for the subject.

Course Learning Objectives (CLO)

The students will be able to:

1. Predict material failure for any combination of applied stresses.
2. Estimate failure conditions of a structure.
3. Determine the stress intensity factor for simple components of simple geometry.
4. Predict the likelihood of failure of a structure containing a defect/flaw.

Text Books

1. Gdoutos, E.E., "Fracture Mechanics: An Introduction", 2nd Ed., Springer. 2005.
2. Broek, D., "Elementary Engineering Fracture Mechanics", 3rd Ed., Springer. 1982.
3. Kumar, P., "Elements of Fracture Mechanics", Wheeler Publishing. 1999.
4. Anderson, T. L., "Fracture Mechanics: Fundamentals and Applications", 3rd Ed., CRC Press. 2005.